

DEFINITION

THE DFT IS NICE AND ALL, BUT IT HAS ONE MAJOR "PROBLEM": IT IS CONTINUOUS. THIS IS A PROBLEM BECAUSE DSP MUST BE ALSO PERFORMED ON COMPUTERS AND $X(e^{j\omega})$ CANNOT BE COMPUTED FOR A CONTINUOUS VARIABLE ω . THE DFT IS THE SOLUTION TO THIS PROBLEM. IT IS ESSENTIALLY A SAMPLED VERSION OF THE DTFT WHICH APPROXIMATES IT.

THE DTFT IS:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$$

↑ CONTINUOUS OUTPUT
↑ DISCRETE INPUT

SINCE WE KNOW IT IS PERIODIC, WE CAN ONLY CONSIDER THE INTERVAL $[0, 2\pi)$ WHICH WE DIVIDE (SAMPLE) IN N ELEMENTS (INDEXED BY k) or $[-\pi, \pi)$

$$\omega = \frac{2\pi}{N} k$$

WITH THIS SUBSTITUTION WE CAN DEFINE THE DFT AS:

$$X[k] := X(e^{j\omega}) \Big|_{\omega = \frac{2\pi}{N} k} = \sum_{n=-\infty}^{+\infty} x[n] e^{-j \frac{2\pi}{N} k n}, \quad k \in [0, N-1]$$

↑ DISCRETE OUTPUT
↑ DISCRETE INPUT

WHICH IS ALSO PERIODIC, SINCE IT IS THE SAMPLED VERSION OF A PERIODIC SIGNAL

INVERSE

$$\tilde{x}[n] := \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi}{N} k n}, \quad k \in [0, N-1]$$

$\tilde{x}[n] \neq x[n]$ BECAUSE IT'S AN APPROXIMATION. LET'S ANALYZE THE DIFFERENCE.

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j \frac{2\pi}{N} k}) e^{j \frac{2\pi}{N} k n}, \quad X(e^{j \frac{2\pi}{N} k}) = \sum_{m=-\infty}^{+\infty} x[m] e^{-j \frac{2\pi}{N} k m}$$

↑ DFT
↑ DFT with $\omega = \frac{2\pi}{N} k$

$$\rightarrow \tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \sum_{m=-\infty}^{+\infty} x[m] e^{-j \frac{2\pi}{N} k m} e^{j \frac{2\pi}{N} k n} = \frac{1}{N} \sum_{k=0}^{N-1} \sum_{m=-\infty}^{+\infty} x[m] e^{j \frac{2\pi}{N} k (n-m)} = \sum_{m=-\infty}^{+\infty} x[m] \underbrace{\frac{1}{N} \sum_{k=0}^{N-1} e^{j \frac{2\pi}{N} k (n-m)}}_{P[n-m]}$$

WE NOW DEFINE

$$P[n] := \frac{1}{N} \sum_{k=0}^{N-1} e^{j \frac{2\pi}{N} k n}$$

SO THAT

$$\tilde{x}[n] = \sum_{m=-\infty}^{+\infty} x[m] p[n-m] = x[n] * p[n]$$

WHICH IS A SIMPLE CONVOLUTION. TO UNDERSTAND WHAT THIS IS WE NEED TO UNDERSTAND WHAT $p[n]$ IS AND TO DO THIS WE IDENTIFY TWO SEPARATE CASES:

CASE 1)

$n = \alpha N \rightarrow p[n] = \frac{1}{N} \sum_{k=0}^{N-1} e^{j \frac{2\pi \alpha N}{N} k} = \frac{1}{N} \cdot N = 1 \Rightarrow p[n] = 1 \text{ if } n = \alpha N, \alpha \in \mathbb{Z}$

n IS A MULTIPLE OF N

$e^{j 2\pi \alpha k}, \alpha, k \in \mathbb{Z} = e^{N j 2\pi}, N \in \mathbb{Z} \rightarrow \text{PERIODIC} = 1$

CASE 2)

$n = \alpha N + r \rightarrow p[n] = \frac{1}{N} \sum_{k=0}^{N-1} e^{j \frac{2\pi (\alpha N + r)}{N} k} = \frac{1}{N} \sum_{k=0}^{N-1} e^{j \frac{2\pi \alpha N}{N} k} e^{j \frac{2\pi r}{N} k} = \frac{1}{N} \sum_{k=0}^{N-1} \underbrace{e^{j 2\pi \alpha k}}_{=1} e^{j \frac{2\pi r}{N} k}$

ON OTHER CASES, n IS NOT A MULTIPLE OF N

$r \in \mathbb{N} \cap [1, N-1]$

$$= \frac{1}{N} \sum_{k=0}^{N-1} e^{j \frac{2\pi r}{N} k} = \frac{1}{N} \sum_{k=0}^{N-1} \left(e^{j \frac{2\pi r}{N}} \right)^k = \frac{1}{N} \frac{1 - e^{j \frac{2\pi r N}{N}}}{1 - e^{j \frac{2\pi r}{N}}} = \frac{1 - 1}{N(1 - e^{j \frac{2\pi r}{N}})} = 0$$

$\sum_{i=a}^{b-1} x^i = \frac{x^a - x^b}{1 - x}$

AND THEREFORE

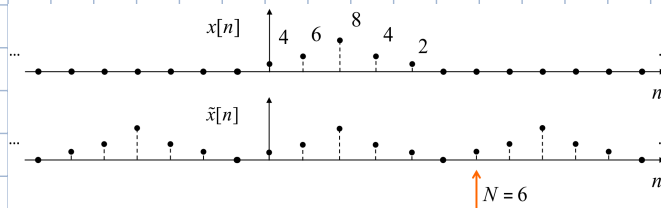
$$p[n] = \begin{cases} 1 & n = \alpha N \\ 0 & \text{OTHERWISE} \end{cases} = \sum_{\alpha=-\infty}^{+\infty} \delta[n - \alpha N]$$

SAME CONCEPT AS SHANNON:
WHEN SAMPLING, SPECTRUM IS REPEATED
EVERY MULTIPLE OF THE SAMPLING FREQUENCY

$$\tilde{x}[n] = x[n] * \sum_{\alpha=-\infty}^{\infty} \delta[n - \alpha N] = \sum_{\alpha=-\infty}^{\infty} x[n - \alpha N]$$

WHICH MEANS THAT THE IDFT RETURNS MULTIPLE REPLICAS OF THE ORIGINAL SIGNAL, WHICH ARE PERIODIC OF PERIOD N .

THIS IS DUE TO THE FACT THAT WE ARE PROJECTING A SIGNAL ONTO A PERIODIC CIRCLE?



! IF THE ORIGINAL SEQUENCE IS OF FINITE LENGTH AND WE TAKE A SUFFICIENT NUMBER OF SAMPLES N FROM ITS DFT, THE ORIGINAL SEQUENCE CAN BE RETRIEVED EXACTLY BY CONSIDERING JUST THE FIRST PERIOD:

$$x[n] = \begin{cases} \tilde{x}[n] & 0 \leq n \leq N-1 \\ 0 & \text{OTHERWISE} \end{cases}$$

MOD NOTATION

$$\langle k \rangle_N = k \bmod N \equiv k \% N$$

← REMAINDER OF $\frac{k}{N}$

CONSIDERING A SIGNAL OF LENGTH N :

$$x[n] = \begin{cases} \neq 0 & 0 \leq n \leq N-1 \\ = 0 & n < 0 \cup n > N-1 \end{cases}$$

← EXISTS ONLY IN $[0, N-1]$

WE CAN ASSOCIATE TO THIS SEQUENCE A PERIODIC SEQUENCE

$$\tilde{x}[n] = \sum_{m=-\infty}^{\infty} x[n-mN] = x[\langle n \rangle_N]$$

ONLY BECAUSE THERE IS NO OVERLAP BETWEEN DELETED REPLICAS

WE CAN APPLY THIS CONCEPT TO THE DFT AS IT IS PERIODIC OF PERIOD N , THUS KEEPING ONLY THE FIRST N VALUES

$$\tilde{X}[k] = X[\langle k \rangle_N]$$

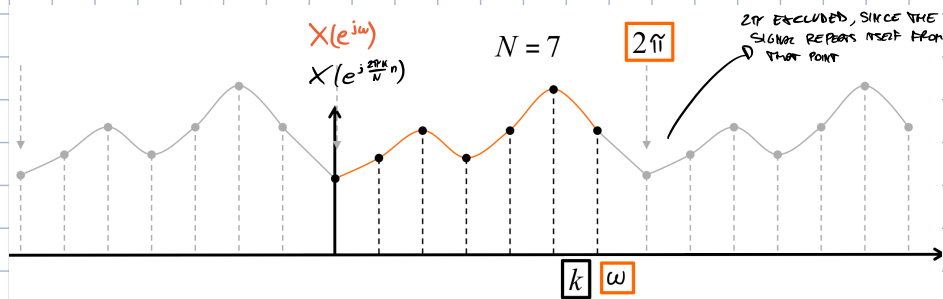
WHICH MEANS THAT WE CAN RE-WRITE DFT AND IDFT AS:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} k n}$$

NOT $[-\infty, \infty]$ ANYMORE
BECAUSE WE ASSUME
 $x[n] = 0$ OUTSIDE OF $[0, N-1]$
↓
CAN NOW BE USED ON A PC

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi}{N} k n}$$

NOT AN APPROXIMATION ANYMORE



SIMPLIFIED NOTATION

BY INTRODUCING $W_N := e^{-j \frac{2\pi}{N}}$ WE OBTAIN

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-nk}$$

! IMPORTANT !

NOTICE THAT THE SIGNS ARE "FLIPPED"

← ENCODED IN W_N

PROOF

THAT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-nk}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-nk} = \frac{1}{N} \sum_{k=0}^{N-1} \sum_{m=0}^{N-1} x[m] W_N^{mk} W_N^{-nk} = \frac{1}{N} \sum_{k=0}^{N-1} \sum_{m=0}^{N-1} x[m] W_N^{(m-n)k} = \sum_{m=0}^{N-1} x[m] \frac{1}{N} \sum_{k=0}^{N-1} W_N^{(m-n)k}$$

LET'S ANALYZE $\frac{1}{N} \sum_{k=0}^{N-1} W_N^{(m-n)k}$:

$$\text{IF } n-m=0 \rightarrow n=m \Rightarrow W_N^0 = 1$$

$$\text{IF } n-m=d (\in \mathbb{Z})$$

$$\frac{1}{N} \sum_{k=0}^{N-1} W_N^{dk} = \frac{1}{N} \sum_{k=0}^{N-1} (W_N^d)^k = \frac{1}{N} \frac{1 - W_N^{dN}}{1 - W_N^d} = \frac{1}{N} \frac{1 - e^{j2\pi d}}{1 - e^{j2\pi \frac{d}{N}}} = 0$$

$\sum_{i=a}^{b-1} x^i = \frac{x^a - x^b}{1-x}$

THEREFORE:

$$\frac{1}{N} \sum_{k=0}^{N-1} W_N^{(m-n)k} = \begin{cases} 1 & m=n \\ 0 & m \neq n \end{cases}$$

SUM '1' EVERY TIME $m=n$ (N TIMES)
AND THEN NORMALIZE BY N

WHICH PROVES THAT

$$x[n] = \dots = \sum_{m=0}^{N-1} x[m] \frac{1}{N} \sum_{k=0}^{N-1} W_N^{(m-n)k} = x[n] \quad \#$$

EX

FIND THE DFT OF $x[n]$

$$x[n] = \delta[n]$$

$$x[n] = 1 \Leftrightarrow n=0$$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{-nk} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{nk}{N}} = 1$$

EX

$$x[n] = \sum_{m=0}^{N-1} \delta[n-m]$$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{-nk} = \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \delta[n-m] W_N^{-nk} = \sum_{n=0}^{N-1} 1 \cdot e^{-j2\pi \frac{nk}{N}} = \frac{1 - e^{-j2\pi k}}{1 - e^{-j2\pi \frac{k}{N}}} = \frac{1 - e^{-j2\pi k}}{1 - e^{-j2\pi \frac{k}{N}}} = 0, k \neq 0$$

IF $k=0 \Rightarrow \frac{0}{0} \Rightarrow$ USE ANOTHER METHOD TO DEFINE IT

$\sum_{n=0}^{N-1} 1 \cdot 1 = N, k=0$

$$X[k] = N \delta[k]$$

EX

$$x[n] = \cos\left(\frac{2\pi r n}{N}\right), r \in \mathbb{Z}$$

$$x[n] = \frac{e^{j\frac{2\pi r n}{N}} + e^{-j\frac{2\pi r n}{N}}}{2} = \frac{W_N^{-rn} + W_N^{rn}}{2}$$

$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk} = \sum_{n=0}^{N-1} \frac{W_N^{-rn} + W_N^{rn}}{2} W_N^{nk} = \frac{1}{2} \sum_{n=0}^{N-1} W_N^{(k-r)n} + W_N^{(k+r)n} = \frac{1}{2} \sum_{n=0}^{N-1} W_N^{(k-r)n} + \frac{1}{2} \sum_{n=0}^{N-1} W_N^{(k+r)n}$$

$$k-r = \alpha N, \alpha \in \mathbb{N}$$

$$\frac{1}{2} \sum_{n=0}^{N-1} W_N^{\alpha N n} = \frac{1}{2} \sum_{n=0}^{N-1} e^{-j2\pi \alpha n} = \frac{N}{2}$$

$$k+r = \alpha N, \alpha \in \mathbb{N}$$

$$\frac{1}{2} \sum_{n=0}^{N-1} W_N^{\alpha N n} = \frac{1}{2} \sum_{n=0}^{N-1} e^{-j2\pi \alpha n} = \frac{N}{2}$$

$$k-r = \alpha N + \beta, \alpha, \beta \in \mathbb{N}$$

$$\frac{1}{2} \sum_{n=0}^{N-1} W_N^{(\alpha N + \beta)n} = \frac{1}{2} \sum_{n=0}^{N-1} \frac{1 - e^{-j\frac{2\pi}{N}(\alpha N + \beta)N}}{1 - e^{-j\frac{2\pi}{N}(\alpha N + \beta)}} = 0$$

$$k+r = \alpha N + \beta, \alpha, \beta \in \mathbb{N}$$

$$\frac{1}{2} \sum_{n=0}^{N-1} W_N^{(\alpha N + \beta)n} = \frac{1}{2} \sum_{n=0}^{N-1} \frac{1 - e^{-j\frac{2\pi}{N}(\alpha N + \beta)N}}{1 - e^{-j\frac{2\pi}{N}(\alpha N + \beta)}} = 0$$

2 CASES WHERE $X[k] \neq 0$:

$$\begin{cases} k = \alpha N + r \\ k = \alpha N - r \end{cases}$$

POINTS WHERE $X[k] \neq 0$ ($= \frac{N}{2}$)

$$r > 0, r \in [0, N-1] \Rightarrow \begin{cases} \alpha = 0 \Rightarrow k = r, r > 0 \\ \alpha \neq 0 \Rightarrow k = N - r, r > 0 \end{cases} \rightarrow X[k] = \frac{N}{2} \delta[k-r] + \frac{N}{2} \delta[k-N+r], r > 0$$

EVEN IF $k=1$, $k=N-r$, BUT N 'S PRESENT THEREFORE IT DOES NOT MATTER

$$r < 0, r \in [-N+1, 0) \Rightarrow \begin{cases} \alpha = 0 \Rightarrow k = -r, r < 0 \\ \alpha \neq 0 \Rightarrow k = N + r, r < 0 \end{cases} \rightarrow X[k] = \frac{N}{2} \delta[k+r] + \frac{N}{2} \delta[k-N-r], r < 0$$

$$X[k] = \frac{N}{2} [\delta[k-r] + \delta[k-N+r] + \delta[k+r] + \delta[k-N-r]]$$

ZERO PADDING

UP TO THIS POINT WE HAVE SEEN THAT IF WE HAVE A SIGNAL $x[n]$ WITH $N=7$ SAMPLES, THE DFT WILL HAVE THE SAME NUMBER OF SAMPLES.

WHAT IF WE WANT MORE SAMPLES IN THE DFT / INCREASE THE SPECTRAL DENSITY?

TO ACHIEVE THIS WE MUST HAVE MORE SAMPLES IN THE TIME DOMAIN. THE SAMPLES OF THE ACTUAL SIGNAL ARE GIVEN AND CANNOT BE INCREASED, BUT WE CAN EXPLOIT THE ASSUMPTION THAT

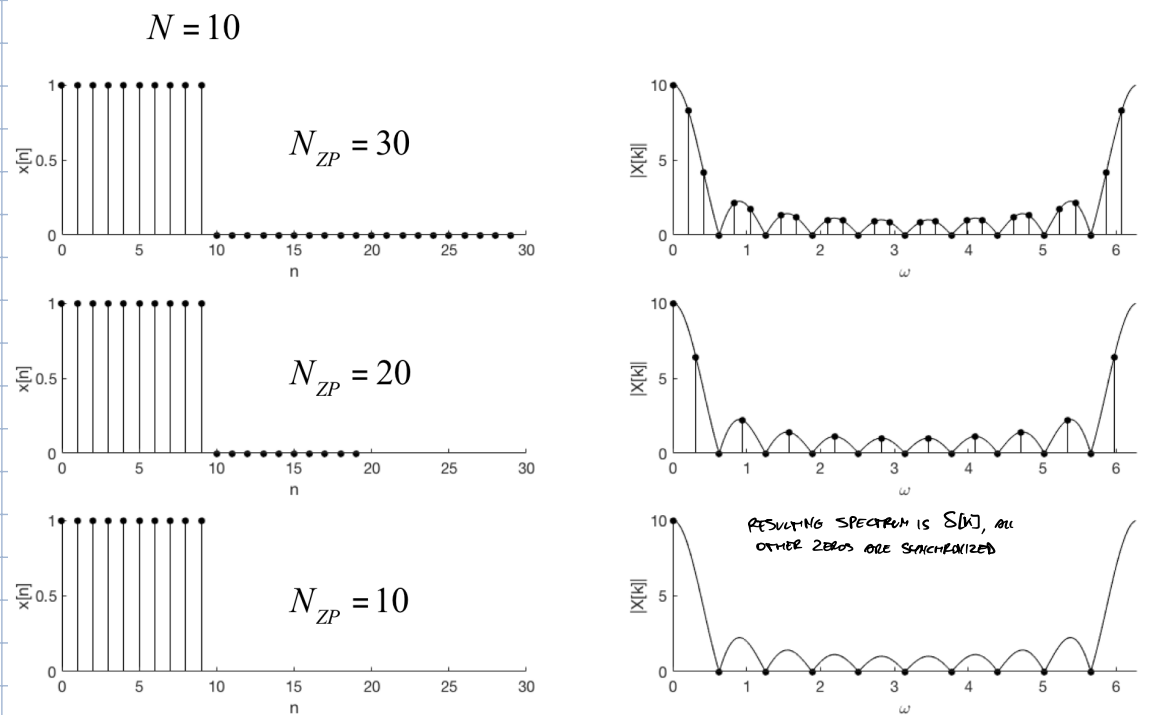
$$x[n] = 0 \quad x < 0 \cup x > N-1$$

AND ACTUALLY INCLUDE ZEROS OUTSIDE (ONLY AFTER) THE ORIGINAL SEQUENCE.

THIS WORKS BECAUSE BY PADDING ZEROS, WE ARE INCREASING THE SPACING BETWEEN REPLICAS (AND THUS DECREASING THE SPACING IN THE FREQUENCY DOMAIN) SUCH THAT THE FOLLOWING STILL HOLDS:

$$\tilde{x}[n] = x[\langle n \rangle_N]$$

$$\Delta\omega = \frac{2\pi}{N}$$



ALIASING IN TIME

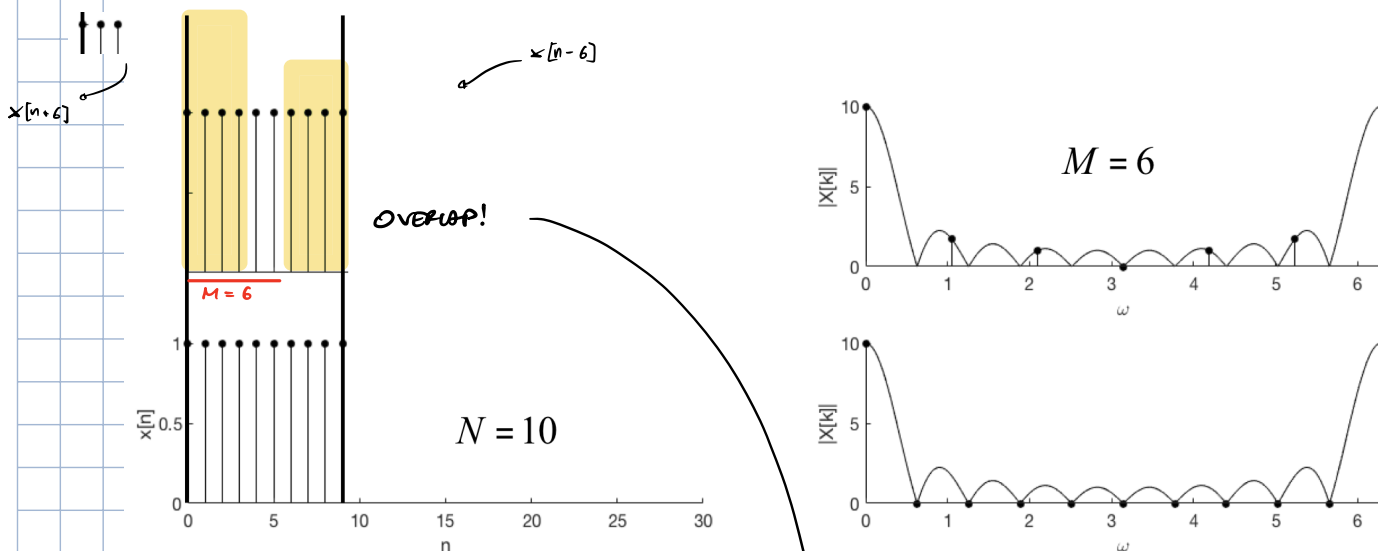
WHAT HAPPENS INSTEAD IF WE ONLY HAVE $M < N$ SAMPLES OF THE DFT?

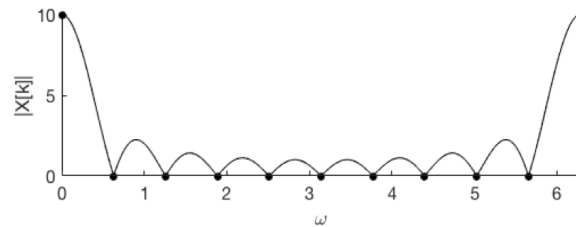
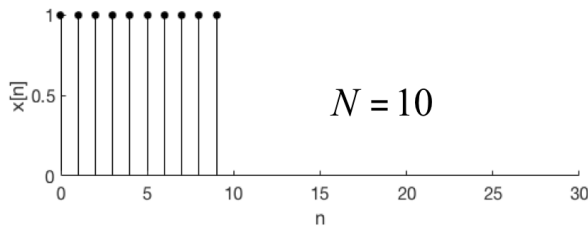
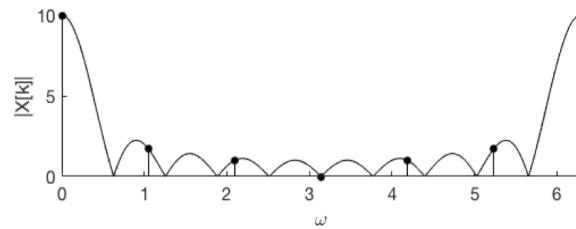
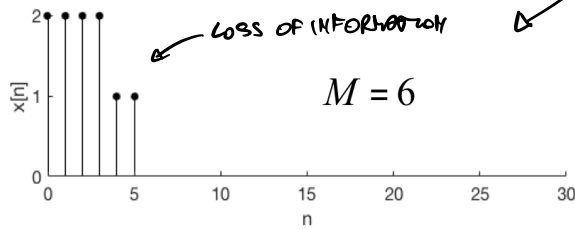
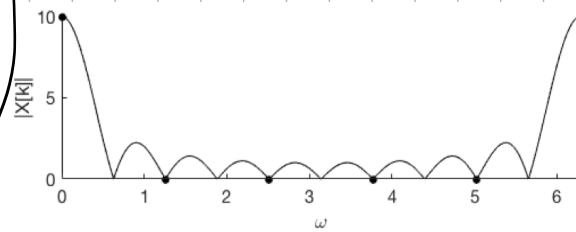
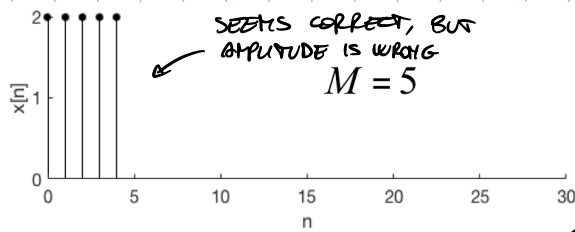
WE CAN STILL ASSOCIATE A PERIODIC SEQUENCE

$$\tilde{x}[n] = \sum_{m=-\infty}^{+\infty} x[n - mM]$$

BUT THE MOD NOTATION ISN'T VALID ANYMORE BECAUSE THERE IS OVERLAP:

$$\tilde{x}[n] \neq x[\langle n \rangle_M]$$





MATRIX REPRESENTATION

THE DFT CAN BE REPRESENTED AS A MATRIX MULTIPLICATION.

$$X[k] = \sum_{n=0}^{N-1} x[n] w_N^{nk}, \quad w_N = e^{-j\frac{2\pi}{N}}$$

$$\begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix} = \begin{bmatrix} w_N^{00} & w_N^{01} & \dots & w_N^{0(N-1)} \\ w_N^{10} & w_N^{11} & \dots & w_N^{1(N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ w_N^{(N-1)0} & w_N^{(N-1)1} & \dots & w_N^{(N-1)(N-1)} \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & w_N^{-1} & w_N^{-2} & \dots & w_N^{-1(N-1)} \\ 1 & w_N^{2} & w_N^{22} & \dots & w_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w_N^{(N-1)} & w_N^{(N-1)2} & \dots & w_N^{(N-1)^2} \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ \vdots \\ x[N-1] \end{bmatrix}$$

$$X = M_N x$$

EX

$N=4$

$$w_N^{ab} = w_N^{(ab)_N}$$

BECAUSE OF $e^{-j\frac{2\pi}{N}ab}$ PERIODICITY

FEW VALUES NEED TO BE COPIED

$$w_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^{-1} & w_4^{-2} & w_4^{-3} \\ 1 & w_4^{2} & w_4^{22} & w_4^{23} \\ 1 & w_4^{3} & w_4^{32} & w_4^{33} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^{-1} & w_4^{-2} & w_4^{-3} \\ 1 & w_4^2 & w_4^4 & w_4^6 \\ 1 & w_4^3 & w_4^6 & w_4^9 \end{bmatrix} \Downarrow = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^{-1} & w_4^2 & w_4^3 \\ 1 & w_4^2 & w_4^0 & w_4^2 \\ 1 & w_4^3 & w_4^2 & w_4^1 \end{bmatrix} =$$

MATRIX IS (OBVIOUSLY) INVERTIBLE

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$W_4^1 = e^{-j\frac{2\pi}{4}} = e^{-j\frac{\pi}{2}} = -j$$

$$W_4^2 = e^{-j\pi} = -1$$

$$W_4^3 = e^{-j\frac{3\pi}{2}} = j$$

"-" MEANS ROTATION IS COUNTERCLOCKWISE AND $\frac{\pi}{2} = 90^\circ \Rightarrow \textcircled{1}$

▽ SIMPLE TRICK: JUST THINK ABOUT HOW MANY STEPS OF $\frac{2\pi}{N}$ ($\frac{\pi}{2}$, $N=4$) WE ARE DOING ON THE COMPLEX PLANE

$$W_4^1 \rightarrow \textcircled{1} \quad \text{! COUNTERCLOCKWISE!}$$

$$W_4^2 \rightarrow \textcircled{2}$$

$$W_4^3 \rightarrow \textcircled{3}$$

$$W_4^0 = W_4^4 \rightarrow \textcircled{0}$$

INVERSE

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-nk}$$

$$\begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} = \frac{1}{N} \begin{bmatrix} W_N^{-00} & W_N^{-01} & W_N^{-02} & \dots & W_N^{-0(N-1)} \\ W_N^{-10} & W_N^{-11} & W_N^{-12} & \dots & W_N^{-1(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ W_N^{-(N-1)0} & W_N^{-(N-1)1} & W_N^{-(N-1)2} & \dots & W_N^{-(N-1)(N-1)} \end{bmatrix} \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix}$$

DON'T FORGET THE FUCKING COEFFICIENT

$$x = M_N^{-1} X$$

EX

$$N=4, \quad x[n] = [0 \ 1 \ 2 \ 3]$$

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 0 + 1 \cdot 1 + 1 \cdot 2 + 1 \cdot 3 \\ 1 \cdot 0 - j \cdot 1 - 1 \cdot 2 + j \cdot 3 \\ 1 \cdot 0 - 1 \cdot 1 + 1 \cdot 2 - 1 \cdot 3 \\ 1 \cdot 0 + j \cdot 1 - 1 \cdot 2 - j \cdot 3 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 + j2 \\ -2 \\ -2 - j2 \end{bmatrix}$$

$$W_4^{-1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -2 & -3 \\ 1 & -2 & -0 & -2 \\ 1 & -3 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

$$x = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 6 \\ -2 + j2 \\ -2 \\ -2 - j2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 6 & -2 + j2 & -2 & -2 - j2 \\ 6 & -j2 + j^2 2 + 2 + j^2 2 + j^2 2 \\ 6 & -2 - j2 & -2 + 2 + j2 \\ 6 & -j2 - j^2 2 + 2 & -j2 - j^2 2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 0 \\ 4 \\ 8 \\ 12 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

OF $\frac{\pi}{2}$ STEPS

$$e^{-j\frac{2\pi}{4}} = e^{-j\frac{\pi}{2}}$$

PROPERTIES

LINEARITY

THE DFT OBEYS LINEARITY, BUT UNDER ONE CONDITION: IF TWO SIGNALS HAVE DIFFERENT DURATIONS, THE NUMBER OF SAMPLES MUST BE AT LEAST THE MAXIMUM OF THE LENGTHS OF THE TWO (ZERO-PAD THE SHORTEST ONE (AT LEAST)).

$$\left. \begin{array}{l} x_1[n] \xrightarrow{\text{DFT}} X_1[k] \\ x_2[n] \xrightarrow{\text{DFT}} X_2[k] \end{array} \right\} \quad x_3[n] = A x_1[n] + B x_2[n] \xrightarrow{\text{DFT}} X_3[k] = A X_1[k] + B X_2[k]$$

↑

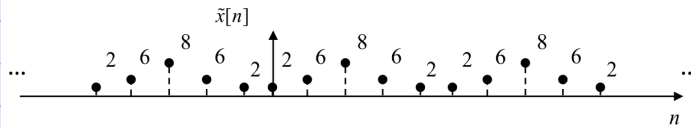
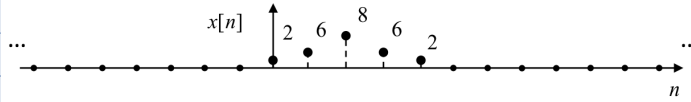
$$N_3 \geq \max\{N_1, N_2\}$$

CIRCULAR REPRESENTATION

AS WE'VE SEEN, THE DFT RETURNS AN INFINITE NUMBER OF REPLICAS OF THE ORIGINAL SEQUENCE.

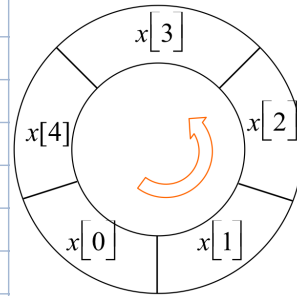
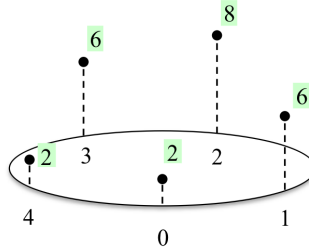
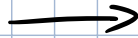
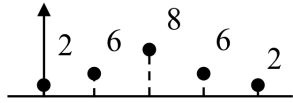
WE ARE ONLY INTERESTED IN ONE REPLIC, THEREFORE AN ALTERNATIVE REPRESENTATION WE CAN USE IS THE CIRCULAR ONE, WHICH ESSENTIALLY REPRESENTS THE MOD OF THE ORIGINAL SEQUENCE.

$$x[n] = 0 \quad x < 0 \cup x \geq N$$



$$\tilde{x}[n] = \sum_{m=-\infty}^{+\infty} x[n-mN] = x[\langle n \rangle_N]$$

PERIODIC VERSION OF $x[n]$ WITH NO OVERLAP



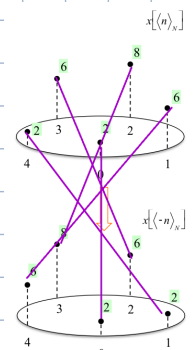
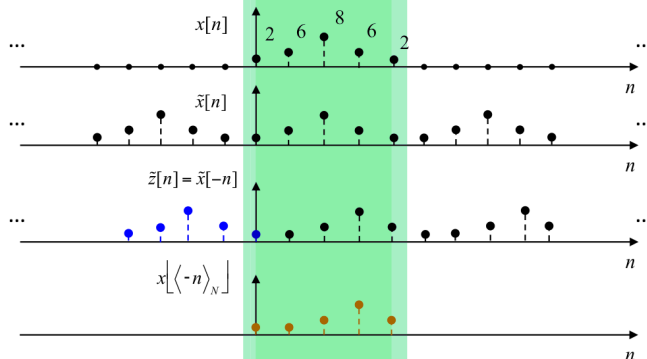
THIS REPRESENTATION IS VERY USEFUL WHEN THE PROPERTIES OF CIRCULAR CONVOLUTION ARE CONSIDERED. IN PARTICULAR IT ALLOWS US TO DEFINE A CONVOLUTION THEOREM THAT IS ANALOG TO THAT OF THE CONTINUOUS TRANSFORMS. THE CIRCULAR CONVOLUTION IS ALSO COMPUTATIONALLY MORE EFFICIENT, IF THE APPROPRIATE ALGORITHMS ARE USED (SPOILER: FFT)

$$x[n] \circledast_N h[n] \xLeftrightarrow{\text{DFT}} X[k] H[k]$$

CIRCULAR FOLDING

LIKE CLASSICAL "PBC-MAX" MODULO, BUT GOING LEFT

$$z[n] = x[\langle -n \rangle_N] = \begin{cases} x[0] & n=0 \\ x[N-n] & 0 < n < N \end{cases}$$

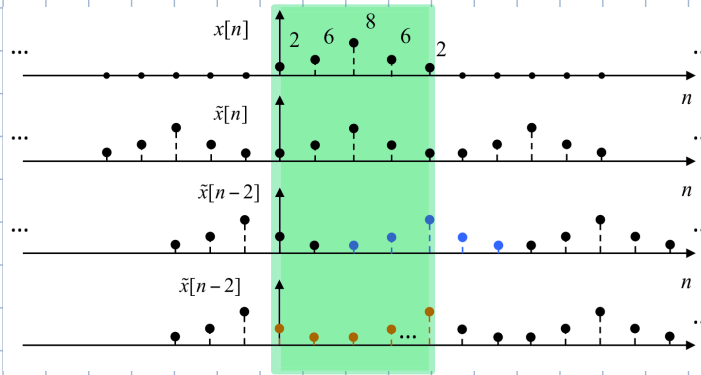


THE CIRCULAR FOLDING HAS THE FOLLOWING PROPERTIES:

$$x_1[n] \xLeftrightarrow{\text{DFT}} X_1[k] \longleftrightarrow x_1[\langle -n \rangle_N] \xLeftrightarrow{\text{DFT}} X_1[\langle -k \rangle_N] = \begin{cases} X_1[0] & k=0 \\ X_1[N-k] & 0 < k < N \end{cases}$$

CIRCULAR TIME SHIFT

$$x_1[n] \xLeftrightarrow{\text{DFT}} X_1[k] \longleftrightarrow x_1[\langle n-m \rangle_N] \xLeftrightarrow{\text{DFT}} X_1[k] e^{-j \frac{2\pi k m}{N}}, \quad 0 \leq n < N$$



CIRCULAR FREQUENCY SHIFT

$$x_1[n] \xLeftrightarrow{\text{DFT}} X[k] \longleftrightarrow x_1[n] e^{j \frac{2\pi k_0 n}{N}} \xLeftrightarrow{\text{DFT}} X[\langle k-k_0 \rangle_N], \quad 0 \leq k < N$$

CIRCULAR CONVOLUTION

$$\left. \begin{array}{l} x[n] \xLeftrightarrow{\text{DFT}} X[k] \\ h[n] \xLeftrightarrow{\text{DFT}} H[k] \end{array} \right\} \quad y[n] = x[n] \circledast_n h[n] = \sum_{k=-\infty}^{+\infty} x[k] h[\langle n-k \rangle_N] = \sum_{k=0}^{N-1} x[k] h[\langle n-k \rangle_N]$$

$$\xrightarrow{\text{DFT}} X[k] H[k]$$

$$\begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[N-1] \end{bmatrix} = \begin{bmatrix} h[0] & h[N-1] & h[N-2] & \cdots & h[1] \\ h[1] & h[0] & h[N-1] & \cdots & h[2] \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ h[N-1] & h[N-2] & h[N-3] & \cdots & h[0] \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}$$

CIRCULAR MATRIX

PROOF:

$$X[k] = \sum_{m=0}^{N-1} x[m] W_N^{km}, \quad H[k] = \sum_{\ell=0}^{N-1} h[\ell] W_N^{k\ell}, \quad Y[k] = X[k] H[k], \quad y[n] = ?$$

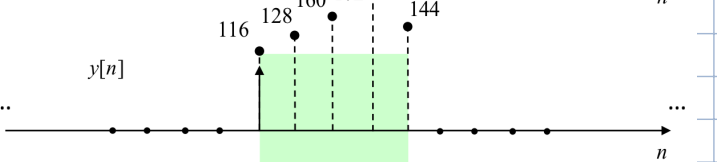
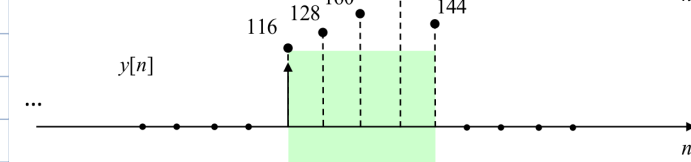
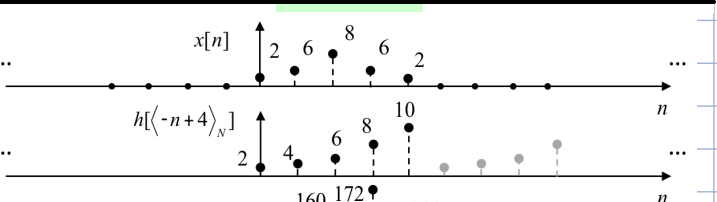
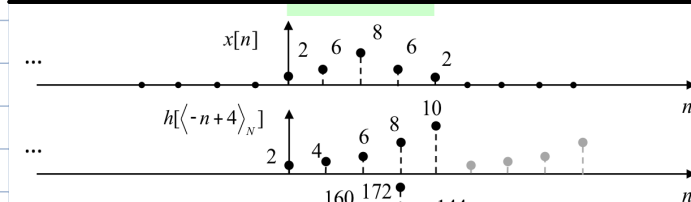
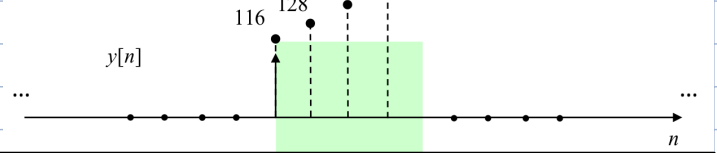
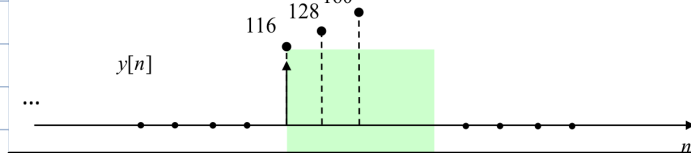
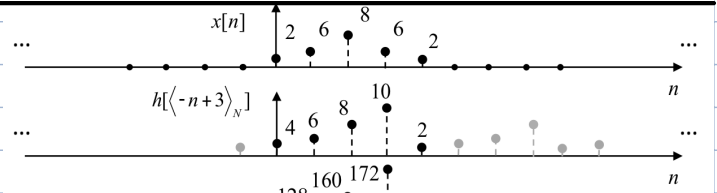
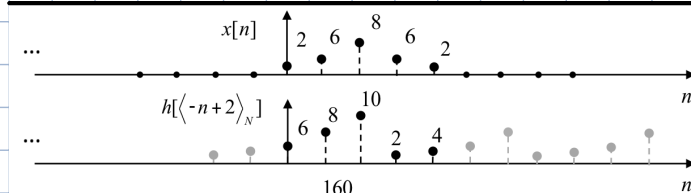
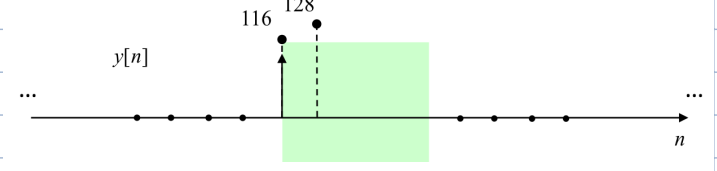
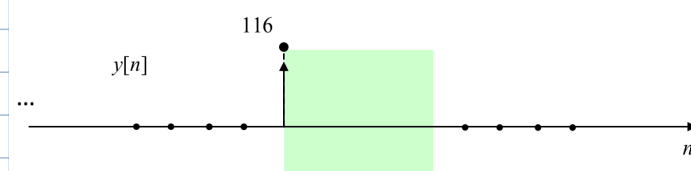
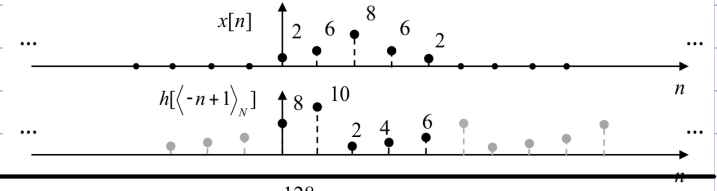
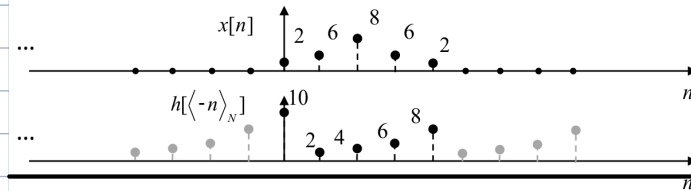
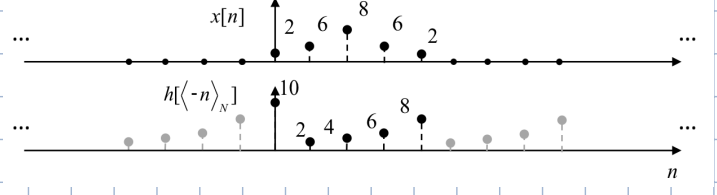
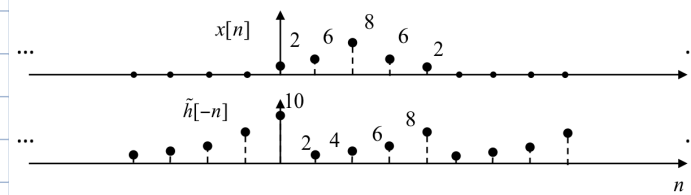
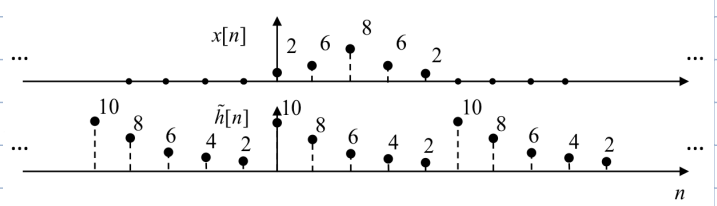
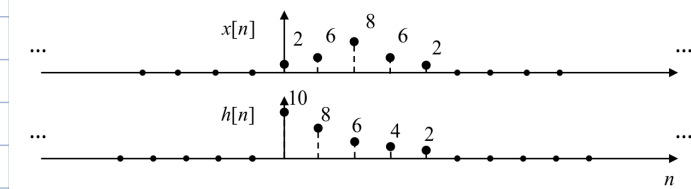
$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} Y[k] W_N^{-kn} = \frac{1}{N} \sum_{k=0}^{N-1} X[k] H[k] W_N^{-kn} = \frac{1}{N} \sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] W_N^{km} \sum_{\ell=0}^{N-1} h[\ell] W_N^{k\ell} \right) W_N^{-kn} =$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] \sum_{\ell=0}^{N-1} h[\ell] \right) W_N^{(m+\ell-n)} = \sum_{m=0}^{N-1} x[m] \sum_{\ell=0}^{N-1} h[\ell] \frac{1}{N} \sum_{k=0}^{N-1} W_N^{(m+\ell-n)}$$

DUE TO PERIODICITY, $h[k] = 0$

$$\frac{1}{N} \sum_{k=0}^{N-1} W_N^{(m+\ell-n)} = \begin{cases} 1, & m+\ell-n = dN \rightarrow m = n-\ell + dN \rightarrow m = \langle n-\ell \rangle_N \\ 0, & \text{OTHERWISE} \end{cases} \quad \text{why?}$$

$$= \sum_{\ell=0}^{N-1} h[\ell] x[\langle n-\ell \rangle_N] \quad \#$$



Ex

$$N=4$$

$$Y[n] = \sum_{m=0}^3 x[m] h[(n-m)_3] = x[n] \circledast_n h[n] = h[n] \circledast_n x[n] \xLeftrightarrow{DFT} X[k] H[k]$$

$$\begin{bmatrix} Y[0] \\ Y[1] \\ Y[2] \\ Y[3] \end{bmatrix} = \begin{bmatrix} h[0] & h[3] & h[2] & h[1] \\ h[1] & h[0] & h[3] & h[2] \\ h[2] & h[1] & h[0] & h[3] \\ h[3] & h[2] & h[1] & h[0] \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

CIRCULAR
MATRIX

Ex

$$N=4$$

$$\left. \begin{array}{l} x[n] = [0 \ 1 \ 2 \ 3] \\ h[n] = [1 \ 0 \ 1 \ 1] \end{array} \right\} Y[n] = x[n] \circledast_n h[n]$$

USING CIRCULAR CONVOLUTION:

POPULATE STARTING FROM ZERO
AND GOING LEFT (WITH MODULO)

$$\begin{bmatrix} Y[0] \\ Y[1] \\ Y[2] \\ Y[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 5 \\ 4 \end{bmatrix}$$

USING DFT:

$$X = M_4 x, \quad H = M_4 h, \quad Y = X[k] H[k], \quad y = M_4^{-1} Y$$

$$W_N = e^{-j \frac{2\pi}{N}} = e^{-j \frac{\pi}{2}}$$

$$W_N^{-1} = e^{j \frac{\pi}{2}}$$

$$M_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$M_4^{-1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ -j-2+3j \\ -1+2-3 \\ j-2-3j \end{bmatrix} = \begin{bmatrix} 6 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix}$$

$$\begin{bmatrix} H[0] \\ H[1] \\ H[2] \\ H[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1-1+j \\ 1+1-1 \\ 1-1-j \end{bmatrix} = \begin{bmatrix} 3 \\ j \\ 1 \\ -j \end{bmatrix}$$

$$\begin{bmatrix} Y[0] \\ Y[1] \\ Y[2] \\ Y[3] \end{bmatrix} = \begin{bmatrix} 6 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix} \begin{bmatrix} 3 \\ j \\ 1 \\ -j \end{bmatrix} = \begin{bmatrix} 18 \\ -2-2j \\ -2 \\ -2+2j \end{bmatrix}$$

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 18 \\ -2-2j \\ -2 \\ -2+2j \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 18 & -2 & -2j & -2 & -2+2j \\ 18 & -2j & +2 & +2 & +2j & +2 \\ 18 & +2 & +2j & -2 & +2 & -2j \\ 18 & +2j & -2 & +2 & -2j & -2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 12 \\ 24 \\ 20 \\ 16 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 5 \\ 4 \end{bmatrix}$$

LINEAR VS. CIRCULAR CONVOLUTION

IS THE CIRCULAR CONVOLUTION ALWAYS EQUIVALENT TO THE LINEAR CONVOLUTION? SPOILER: NO
IT IS EQUIVALENT ONLY IF A PARTICULAR CONDITION IS SATISFIED.

IN GENERAL THE LINEAR CONVOLUTION IS THE FOLLOWING:

$$\left. \begin{array}{l} x[n] \xleftrightarrow{\text{DFT}} X(e^{j\omega}) \\ h[n] \xleftrightarrow{\text{DFT}} H(e^{j\omega}) \end{array} \right\} y[n] = \sum_{m=-\infty}^{\infty} x[n] h[n-m] = x[n] * h[n] \xleftrightarrow{\text{DFT}} Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$$

LET

$$\left. \begin{array}{l} \text{LEN}\{x[n]\} = N_1 \\ \text{LEN}\{h[n]\} = N_2 \end{array} \right\} \text{LEN}\{y[n]\} = N_1 + N_2 - 1$$

IF WE CHOOSE $N = \max\{N_1, N_2\}$ AND COMPUTE AN N POINT CIRCULAR CONVOLUTION, WE GET AN N POINT SEQUENCE.

THIS SEQUENCE IS THE OUTPUT OF THE CIRCULAR CONVOLUTION AND IS OBVIOUSLY DIFFERENT FROM THE OUTPUT OF THE LINEAR CONVOLUTION (WHICH IS $y[n]$), UNLESS BOTH $x[n]$ AND $h[n]$ ARE EXTENDED TO $N_2 + N_1 - 1$ BY PADDING ZEROS.

IF THIS CONDITION IS SATISFIED, THE CIRCULAR CONVOLUTION IS IDENTICAL TO THE LINEAR CONVOLUTION.

CONCEPT EXPLANATION: WE ARE ADDING ENOUGH ZEROS SUCH THAT THE TWO SEQUENCES NEVER OVERLAP MORE THAN ONCE PER PASS.

AND THE CONVOLUTION PASS IS EXACTLY THE SAME AS THE NORMAL ONE AND ONLY THE PADDED ZEROS ARE "LOOPING" AND RETURNING TO THE START OF THE SEQUENCE

$N_1 = 5$ a b c d e 0 0 0 0 0 0 ∇ FOLDING NOT PERFORMED; JUST FOR CONCEPT
 $N_2 = 7$ A B C D E F G 0 0 0 0 0

$$5+7-1=11$$

a b c d e 0 0 0 0 0 0 a b c d e 0 0 0 0 0 0 a b c d e 0 0 0 0 0 0
A B C D E F G 0 0 0 0 0 0 0 0 A B C D E F G G 0 0 0 0 A B C D E F

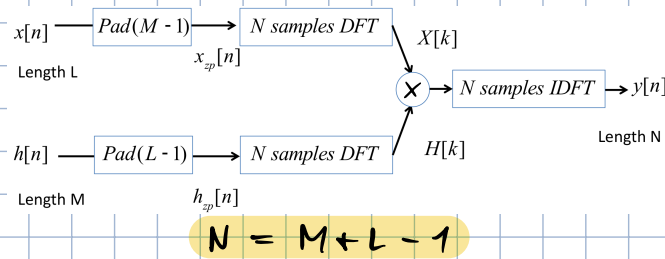
IF IT HAD ONE
ZERO LESS

a b c d e 0 0 0 0 0
G 0 0 0 A B C D E F

COMPUTATION

THIS RESULT IS QUITE REMARKABLE, AS IT TELLS US THAT WE CAN COMPUTE A LINEAR CONVOLUTION BY PERFORMING THE DFT, WHICH IS COMPUTATIONALLY MORE EFFICIENT.

THIS IS CLEARLY TRUE ONLY IF THE LENGTH CONSTRAINT IS SATISFIED.



LINEAR CONVOLUTION IS $\Theta(N^2)$, CIRCULAR CONVOLUTION IS $\Theta(N \log N)$ ^{WITH FFT}

EX

$$x[n] = [0 \ 1 \ 2 \ 3]$$

$$h[n] = [1 \ 0 \ 1 \ 1]$$

WITH LINEAR CONVOLUTION:

$$y[n] = x[n] * h[n] = \sum_{m=-\infty}^{+\infty} x[m] h[n-m]$$

$$y[0] = \sum_{m=-\infty}^{+\infty} x[m] h[-m] = \begin{matrix} [0 & 1 & 2 & 3] \\ [1 & 1 & 0 & 1] \end{matrix} = 0 \cdot 1 = 0$$

$$y[1] = \sum_{m=-\infty}^{+\infty} x[m] h[-m+1] = \begin{matrix} [0 & 1 & 2 & 3] \\ [1 & 1 & 0 & 1] \end{matrix} = 1$$

$$y[2] = \sum_{m=-\infty}^{+\infty} x[m] h[-m+2] = \begin{matrix} [0 & 1 & 2 & 3] \\ [1 & 1 & 0 & 1] \end{matrix} = 2$$

$$y[3] = \sum_{m=-\infty}^{+\infty} x[m] h[-m+3] = \begin{matrix} [0 & 1 & 2 & 3] \\ [1 & 1 & 0 & 1] \end{matrix} = 4$$

$$y[4] = \sum_{m=-\infty}^{+\infty} x[m] h[-m+4] = \begin{matrix} [0 & 1 & 2 & 3] \\ [1 & 1 & 0 & 1] \end{matrix} = 3$$

$$y[5] = \sum_{m=-\infty}^{+\infty} x[m] h[-m+5] = \begin{matrix} [0 & 1 & 2 & 3] \\ [1 & 1 & 0 & 1] \end{matrix} = 5$$

$$y[6] = \sum_{m=-\infty}^{+\infty} x[m] h[-m+6] = \begin{matrix} [0 & 1 & 2 & 3] \\ [1 & 1 & 0 & 1] \end{matrix} = 3$$

$$y[n] = [0 \ 1 \ 2 \ 4 \ 3 \ 5 \ 3]$$

WITH CIRCULAR CONVOLUTION, NO PADDING

$$y[n] = x[n] \circledast_3 h[n] = \sum_{m=0}^3 x[m] h[\langle n-m \rangle_3]$$

$$y = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 5 \\ 4 \end{bmatrix}$$

WITH PADDING

$$N_{zp} = 4 + 4 - 1 = 7$$

$$x[n] = \begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 0 & 0 \end{bmatrix}$$

$$h[n] = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 4 \\ 3 \\ 5 \\ 3 \end{bmatrix}$$

PROPERTIES CONTD.

DUALITY

$$x[n] \xleftrightarrow{\text{DFT}} X[k] \longleftrightarrow X[n] = N x[\langle -n \rangle_N]$$

THE SAME "SHAPE" OF THE FREQUENCY SPECTRUM REPLICATED IN THE TIME DOMAIN

PARSEVAL'S THEOREM

$$\sum_{n=0}^N x[n] h^*[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] H^*[k]$$

THE SUM OF THE SAMPLE BY SAMPLE PRODUCTS OF TWO COMPLEX SEQUENCES OF LENGTH N EQUALS THE NORMALIZED SUM OF THE COEFFICIENT BY COEFFICIENT PRODUCTS OF THEIR DFT. —> ESSENTIALLY CONSERVATION OF ENERGY SINCE INFORMATION IS UNCHANGED (BEFORE OF $1/N$)

IT CAN BE USED TO COMPUTE THE ENERGY OF THE SIGNAL

$$E_x = \sum_{n=0}^N |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$$

SYMMETRY PROPERTIES

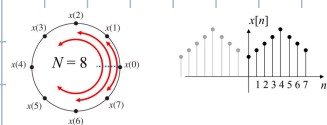
CIRCULAR SYMMETRY OF REAL SEQUENCE

GIVEN A REAL SEQUENCE $x[n]$ OF LENGTH N

$x[n]$ IS CALLED "CIRCULAR EVEN" IF

$$x[n] = x[\langle -n \rangle_N]$$

EVEN SYMMETRY ABOUT 0 ON THE CIRCLE



$x[n]$ IS CALLED "CIRCULAR ODD" IF

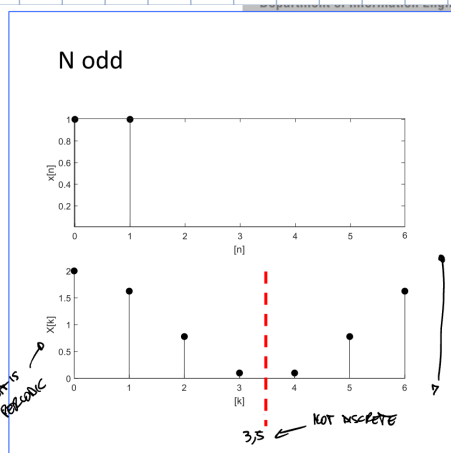
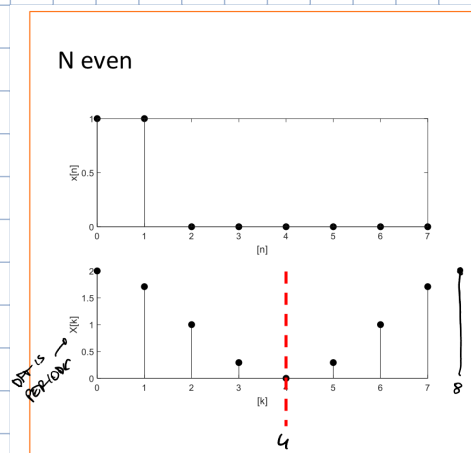
$$x[n] = -x[\langle -n \rangle_N]$$

$$x[n] = x^*[n] \leftrightarrow X^*[\langle -k \rangle_N] = X[k]$$

Sequence $x[n]$	Discrete Fourier Transform $X[k]$
Any real $x[n]$	$X[k] = X^*[\langle -k \rangle_N]$ Complex part is odd, circular symmetric. Real part is even, circular symmetric.
Any real $x[n]$	$X_{Re}[k] = X_{Re}[\langle -k \rangle_N]$
Any real $x[n]$	$X_{Im}[k] = -X_{Im}[\langle -k \rangle_N]$
Any real $x[n]$	$ X[k] = X[\langle -k \rangle_N] $
Any real $x[n]$	$\angle X[k] = -\angle X[\langle -k \rangle_N]$
$x_e[n]$	$X_{Re}[k]$
$x_o[n]$	$jX_{Im}[k]$

$$\left. \begin{array}{l} x = a + jb \\ x^* = a - jb \end{array} \right\}$$

IF N IS ODD, THE AXIS OF SYMMETRY ($N/2$) CORRESPONDS TO A SAMPLE, IF INSTEAD IT IS EVEN, IT LIES IN BETWEEN TWO SAMPLES



CIRCULAR SYMMETRY OF COMPLEX SEQUENCE

GIVEN A COMPLEX SEQUENCE $x[n]$ OF LENGTH N

$$x[n] = x_{cs}[n] + x_{co}[n], \quad 0 \leq n \leq N-1$$

$$\begin{aligned} 2x_{cs}[n] &= a + jb + a - jb = 2a \\ 2x_{co}[n] &= a + jb - a + jb = 2jb \end{aligned}$$

Real part $\rightarrow$
imag. part $\rightarrow$

WHERE:

- CIRCULAR CONJUGATE SYMMETRIC: $X_{cs}[n] = \frac{1}{2} \left[x[\langle -n \rangle_N] + x^*[\langle -n \rangle_N] \right], 0 \leq n \leq N-1$

- CIRCULAR CONJUGATE ANTI-SYMMETRIC: $X_{ca}[n] = \frac{1}{2} \left[x[\langle -n \rangle_N] - x^*[\langle -n \rangle_N] \right], 0 \leq n \leq N-1$

$x[n]$ IS SAID TO BE CONJUGATE SYMMETRIC IF: $x[n] = x^*[\langle -n \rangle_N]$

$x[n]$ IS SAID TO BE CONJUGATE ANTI-SYMMETRIC IF: $x[n] = -x^*[\langle -n \rangle_N]$

CIRCULAR SYMMETRY OF DFT

GIVEN $X[k]$ OF LENGTH N

$$X[k] = X_{cs}[k] + X_{ca}[k], 0 \leq k \leq N-1$$

WHERE:

- CIRCULAR CONJUGATE SYMMETRIC: $X_{cs}[k] = \frac{1}{2} \left[X[\langle -k \rangle_N] + X^*[\langle -k \rangle_N] \right], 0 \leq k \leq N-1$

- CIRCULAR CONJUGATE ANTI-SYMMETRIC: $X_{ca}[k] = \frac{1}{2} \left[X[\langle -k \rangle_N] - X^*[\langle -k \rangle_N] \right], 0 \leq k \leq N-1$

$X[k]$ IS SAID TO BE CONJUGATE SYMMETRIC IF: $X[k] = X^*[\langle -k \rangle_N]$

$X[k]$ IS SAID TO BE CONJUGATE ANTI-SYMMETRIC IF: $X[k] = -X^*[\langle -k \rangle_N]$

Sequence $x[n]$	Discrete Fourier Transform $X[k]$
$x^*[n]$	$X^*[\langle -k \rangle_N]$ DFT OF CONJUGATE IS FOLDED
$x^*[\langle -n \rangle_N]$	$X^*[k]$ DFT OF FOLDED CONJUGATE IS KOTHEL
$\text{Re}\{x[n]\}$	$X_{cs}[k]$ (circular conjugate-symmetric part)
$j\text{Im}\{x[n]\}$	$X_{ca}[k]$ (circular conjugate-antisymmetric part)
$x_{cs}[n]$	$X_{\text{Re}}[k] = \text{Re}\{X[k]\}$
$x_{ca}[n]$	$jX_{\text{Im}}[k] = j\text{Im}\{X[k]\}$